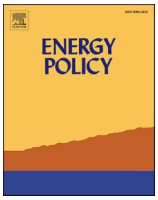
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Greenhouse gas emission forecasts for electrification of space heating in residential homes in the US

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ABSTRACT

This study aims to inform policymakers about the greenhouse gas emission impacts of heat pump deployment in residential homes. Electric heat pumps eliminate direct burning of fossil fuels in homes but result in indirect emissions due to fossil fuels burned for electricity production. This paper presents the first detailed emission forecasts for operating either a heat pump or gas furnace for residential heating over a 15-year period, starting in year 2022 through 2036, in six regions across the US. The study accounted for long-run marginal emissions from electricity generation, emissions from natural gas combustion in homes, and fugitive methane and refrigerant emissions from leaks. The population weighted US average results show emission reductions for a heat pump over furnace to be 38–53% for carbon dioxide, 53–67% for 20-Year global warming potential (GWP), and 44–60% for 100-Year GWP, with reductions increasing over time. The impact of fugitive emissions from the furnace is significantly higher than that of the heat pump. While more energy efficient construction reduces overall emissions for both heating types, the forecasted percent emission reduction for replacement of a gas furnace with heat pump was not impacted by changes in home construction parameters.

1. Introduction

Residential energy use leads to about 20% of the total greenhouse gas (GHG) emissions in the US (Goldstein et al., 2020). The need for decarbonization to limit global warming is driving electrification of end uses in the building sector that have been historically fueled primarily by natural gas (Bailey et al., 2019). The 2015 Residential Energy Consumption Survey reported that approximately half of homes in the US are heated by equipment that relies on combustion of natural gas (U.S. Energy Information Administration, 2015a). Electrification of residential space heating using heat pumps is a potential strategy to reduce GHG emissions associated with natural gas combustion and fugitive methane emissions. While electrification with heat pumps eliminates direct combustion emissions from furnaces on-site, a complete comparison of the GHG emissions must consider time-varying on-site and off-site emissions associated with electricity and natural gas production. Furthermore, the fugitive emissions of refrigerant used in heat pumps should be included in forecasting emissions to provide a complete picture for policy analysis.

Previous work has forecasted that the environmental impact of meeting greater electricity demand is dependent on the carbon intensity of generation sources operating at the margin, which varies by geographic location and by time of day and year (Callaway et al., 2015) (Graff Zivin et al., 2014) (Siler-Evans et al., 2012). A number of studies predicted electrification impacts on GHG emissions in the year 2050 for various geographic regions while considering diverse grid models and heating equipment (Calderon et al., 2019) (Tarroja et al., 2018) (Roux et al., 2016). However, when considering the emissions reduction potential of new heat pump installations, the forecasted carbon-intensity of the electricity generated to operate the heat pump over its life must be considered, where the median service life of a heat pump is estimated at 15 years (American Society of Heating, 2019). Brockway and Delforge (2018) and Tarroja et al. (2018) performed a carbon dioxide (CO_2) emission study on electrification of space and water heating but restricted the analysis to California. Raghaven et al. (2017) studied GHG impact from the electrification of water heating technologies in California while considering transition to new refrigerants and the leaks associated with it.

The current study broadens the research scope to the entire US while considering location specific forecasted hourly climate and grid emissions. Another important factor affecting heat pump adoption and GHG emissions, not considered by the previous studies, is the use of less efficient backup electric resistance heat strips, especially for cold

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Abbreviations

AHU – Air handling unit
BTM – Behind the meter
CO_2 – Carbon dioxide
CO_2-eq – Carbon dioxide equivalent
COP – Coefficient of performance
EIA – Energy Information Administration
EPA – Environmental Protection Agency
GF – Gas furnace
GHG – Greenhouse gas
GWP – Global warming potential
HP – Heat pump
IECC – International Energy Conservation Code
NREL – National Renewable Energy Laboratory
PNNL – Pacific Northwest National Laboratory

Nomenclature

Cap – nominal cooling capacity of the heat pump [kw]
$co2_lrmer_enduse_h$ – hourly emissions per end-use electricity consumed [kg $C0_2$/kWh]
E_h – hourly electricity consumed [kWh]
f_{ac} – fraction of homes with existing air conditioning
f_e – lifetime refrigerant emissions ratio
$m_{CO2_elec_annual}$ – mass of annual CO_2 emissions from electricity consumed [kg $C0_2$]
$m_{Ref_lifetime}$ – mass of refrigerant emissions over 15-year lifetime [kg]
y – year of heat pump installation

climates (Larson et al., 2013). The current study took this into account by sizing the heat pump and electric resistance strip heat capacity using standard industry sizing methods. Furthermore, the current study used real air-source heat pump coefficient of performance (COP) curves over a range of outdoor conditions for accurate electricity consumption predictions, which was lacking in previous studies. Finally, this study accounts for methane and refrigerant emissions, which are important to quantity and consider because of their high global warming potential (GWP) in comparison to CO_2 generated from combustion.

The goal of this work was to develop a framework to compare emissions for heat pumps and gas furnaces to inform heating electrification policies across the US. This work analyzed the expected 15-year operational GHG emission impacts resulting from replacing a residential natural gas furnace with a variable speed heat pump in a single-family home simulated in 99 locations across the US. The analysis considers all known GHG emissions sources associated with gas furnaces and heat pumps: long-run end-use marginal CO_2 emissions from electricity generation, CO_2 emissions from natural gas combustion, and fugitive methane and refrigerant emissions from leaks. Fugitive methane and refrigerant emissions were factored into the analysis by analyzing the CO_2 equivalent (CO_2-eq) of the GWP of these gases on a 20-year and 100-year timescale. The results of the analysis may be used to inform policymakers of the GHG implications of residential heat pump adoption in the US.

The paper includes a methodology section that describes the EnergyPlus residential building model, geographic locations and hourly weather data, and emission sources considered. The methodology is applied to 99 locations across the US and results are aggregated into population-weighted averages for six regions for heating systems installed from 2022 to 2035 and operating for 15-years. Conclusions and policy implications are discussed based on the results.

2. Methodology

Recent work has demonstrated that a detailed analysis based on hourly heat pump demand and grid energy generation mix is critical for accurately evaluating GHG emissions (Neirotti et al., 2020). The total CO_2, 20-year and 100-year GWP emissions were hourly forecasted for an electric variable speed heat pump and a single-stage high efficiency gas furnace used for heating a residential home. The emissions were forecasted for each piece of equipment used over a 15-year operating life, with installation dates ranging from 2022 to 2036 in two-year increments, in 99 cities spread over six regions of the US. Emissions sources analyzed for the heat pump included CO_2 from marginal electricity generation (to operate the heat pump compressor, electric resistance heat strip, and the air handling unit (AHU) fan) and refrigerant leaks associated with an expected increase in refrigerant use (Table 1). Emission sources analyzed for the gas furnace included CO_2 from electricity generation (to operate the AHU fan) and from natural gas (i.e., methane) combustion, and methane leaks from natural gas production and leaks downstream of the residential meter (also called "behind the meter" leakage) (Table 1).

Table 1
Emission sources analyzed for each heating system.

Metric	Source	Heat Pump (HP)	Gas Furnace (GF)
Carbon Dioxide	End-Use Electricity – Air Handler Fan	X	X
Carbon Dioxide	End-Use Electricity – Compressor	X	
Carbon Dioxide	End-Use Electricity – Electric Resistance Heat	X	
Carbon Dioxide	End-Use Natural Gas – Combustion		X
20-Year and 100-Year GWP	Refrigerant Leaks	X	
20-Year and 100-Year GWP	Methane Leaks – Production	X	X
20-Year and 100-Year GWP	Methane Leaks – Downstream of Meter		X

2.1. Residential building model

Selection of a representative residential building model for the US is challenging because heating loads for a specific home will be impacted by building codes and home size, which vary by location and year of construction. In order to estimate the impact due to this variation, two EnergyPlus (U.S. Department of Energy, 2021) models from Pacific Northwest National Laboratory (PNNL) (Pacific Northwest National Laboratory, 2020) representative of a home constructed on a concrete slab to California building codes were used as the basis of the simulations. Two models were chosen, one of an older construction year (2006) and another of a newer construction year (2018), in order to understand the impact of varying home construction types on the GHG emissions from heating technologies. The models were updated to run with EnergyPlus version 9.4 and a newer EnergyPlus object (Construction: FfactorGroundFloor) was used to simulate the temperature of the perimeter of the concrete slab. This object uses ground temperatures from the weather file and the properties of the slab to model the heat exchange from ground to slab to conditioned space. Two mechanical systems were simulated: a gas furnace with single speed air conditioner and a variable speed heat pump with an electric resistance heat strip as a backup heat source.

2.1.1. Geometry, construction, and internal loads

The geometry, construction, and internal loads in the home were specified by the PNNL models used (Pacific Northwest National Laboratory, 2020). The single-family residence model had one zone of 223 m^2 of conditioned space and 111.5 m^2 of unconditioned attic space. The main differences between the 2006 and 2018 construction years were building envelope properties including increased insulation, reduction in effective leakage area (which reduces infiltration), and improved windows (Table 2).

2.1.2. Gas furnace model

A fixed capacity, high efficiency condensing gas furnace based on commercially available equipment (Trane S9X1 (Trane, 2019)) with annual fuel utilization efficiency of 0.96 was modeled. The rated capacity of the gas furnace was set to 29.3 kW, which was selected for simplicity to meet the heating loads in all climates with less than 25 unmet heating hours annually. While the modeled furnace rated capacity is larger than necessary in some climates, the amount of combustion fuel used by the furnace is accurately calculated to meet the heating setpoints in each climate and is the sole basis for all GHG emission prediction. The heating setpoint for the gas furnace was set to 18.3 °C between the hours of 11pm-7am and to 20.0 °C between 7am and 11pm, based on California's 2019 Residential Alternative Calculation Method Reference Manual (California Energy Commission, 2019).

The EnergyPlus simulation outputs used in the analysis were the hourly natural gas consumption by the gas furnace and the electricity used by the air handler fan only during heating mode. The electricity used for cooling was not considered in this analysis.

2.1.3. Heat pump model

A variable speed heat pump was modeled based on manufacturer provided data tables for a commercially available residential variable speed heat pump (Outdoor Unit: Trane Technologies Model 4TWV0036A1, Indoor Unit: Trane Technologies Model TAM9-A0C36V31) with a heating COP of three at the rated condition (outdoor air dry bulb temperature of 8.3 °C and indoor air dry bulb temperature of 21.1 °C) (Fig. 1). The capacity ratio, defined as the delivered capacity divided by the capacity at the rated conditions, increased with increasing compressor speed and outdoor air dry bulb temperature. The model selected the compressor speed inside the bounds of the minimum and maximum speed necessary to provide the required heating capacity and meet the thermostat setpoint. Note that the capacity ratio at the minimum speed in Fig. 1 increases below 5 °C because the manufacturer sets a higher minimum compressor speed as outdoor temperature decreases below this threshold. The heat pump model used an automatic compressor reverse-cycle outdoor coil defrost strategy and contained electric resistance heat strips that operated with the heat pump to meet the heating load when it could not be met by the heat pump alone. This heat pump model is representative of cold climate heat pumps and is capable of heating within the ambient temperature range from −23 °C to 19 °C. Since the COP of the air source heat pump approached 1 at very low outdoor air temperatures, the model switched the heat pump off below −18 °C and used the electric resistance heat strips to provide the required heating.

Table 2
Main construction differences between 2006 and 2018 construction years from PNNL models (Pacific Northwest National Laboratory, 2020).

	Home Vintage - 2006	Home Vintage - 2018
Attic Floor Insulation R-value [m^2·K/W]	4.61	5.51
Exterior Wall Insulation R-value [m^2·K/W]	1.53	2.44
Exterior Door Insulation R-value [m^2·K/W]	0.27	0.55
Occupied Space Effective Leakage Area [m^2]	0.095	0.036
Attic Effective Leakage Area [m^2]	0.037	0.037
Window U-factor [W/(m^2·K)]	3.69	1.82
Window Solar Heat Gain Coefficient	0.3	0.2

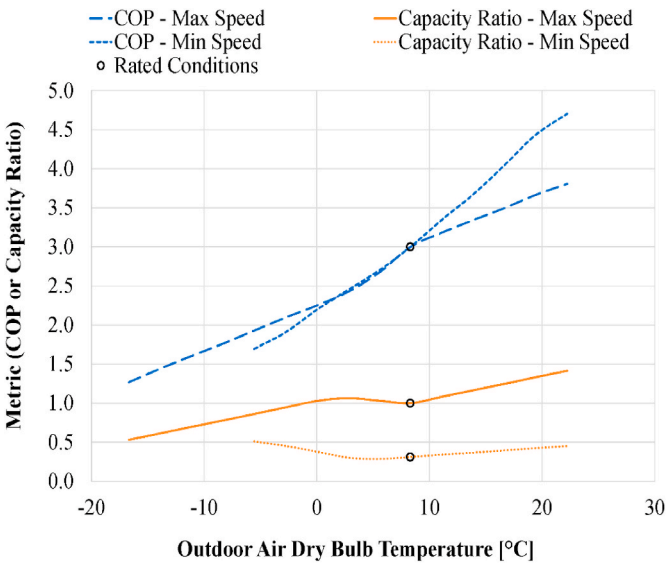


Fig. 1. Heat pump COP and capacity ratio as a function of outdoor air dry bulb temperature. Indoor condition dry bulb temperature was 21.1 °C for the data shown.

The EnergyPlus autosize routine was used to automatically size the heat pump capacity. It is common practice to size systems with a sizing factor between 1.33 and 1.48 to ensure that the load can be met all year (Office of Energy Efficiency and Renewable Energy, 2018). Consistent with industry practice, the autosize routine sized the heat pump capacity to meet 1.4 times the 99.6% design day cooling load. The electric resistance heat strips (COP of 1) were then sized to meet the entire heating load (which would be required when the outdoor air temperature drops below the low-temperature shut-off or in practice during defrost-cycle operation). The heating setpoint for the heat pump was the same as for the home with the gas furnace.

The EnergyPlus simulation outputs used in the analysis were the hourly electricity used by the heat pump compressor during heating mode, heat pump electric resistance heat strips, and air handler fan during heating mode only. The electricity used for air conditioning was not considered in this analysis.

2.2. Geographic locations and weather

Simulations were conducted for weather representative of 99 cities across the Continental US. The largest city was selected from each International Energy Conservation Code (IECC) climate region (Pacific Northwest National Laboratory, 2015) within each state, which resulted in 97 cities simulated (Appendix A). In order to increase resolution of data within California, which is largely in IECC climate region 3, two additional cities (San Jose and Sacramento) were selected. Typical meteorological year 3 (TMY3) weather files in EPW file format were purchased for all 99 locations from White Box Technologies (White Box Technologies, 2015). Weather was assumed to remain the same for each year of the analysis period.

Greenhouse gas emissions results were analyzed by geographic region, where the regions were defined as Pacific, Rocky Mountains, Southwest, Midwest, Northeast, and Southeast (Fig. 2). The results for each region were calculated with a population weighted average based on the population of the county in which each city was located, using data from a July 2019 US Census estimate (United States Census Bureau, 2010).

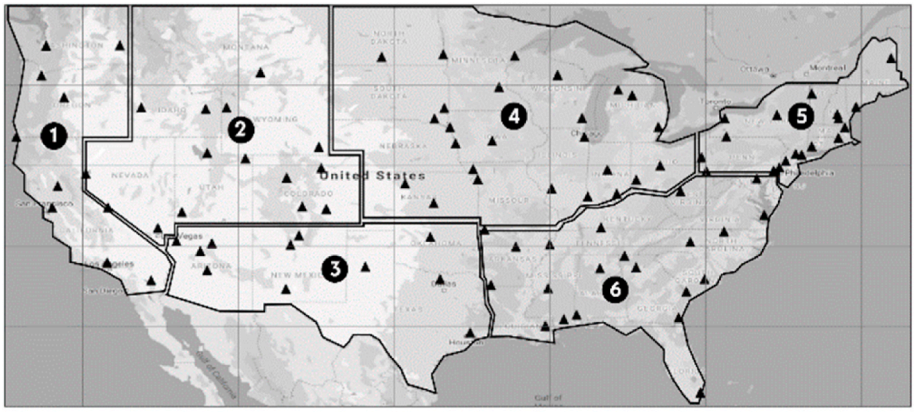


Fig. 2. Regions used for analysis of results. Each city simulated is marked by a black triangle.

2.3. Emission sources

2.3.1. Long-run marginal carbon dioxide emissions from electricity generation

Long-run end-use marginal carbon dioxide emissions for electricity generation from the National Renewable Energy Laboratory's (NREL) Cambium tool (Gagnon et al., 2020) (National Renewable Energy Laboratory, 2020) were applied to forecast the impacts of the electricity consumptions. The Cambium tool provided the hourly cost, emission, and operational data for modeled future scenarios of the US electricity sector in two-year increments (2022–2050). Cambium projections are conservative forecasts based on state level legislation passed as of June 30, 2020. This project used the low-renewable energy cost scenario, which was one of the five scenarios modeled by NREL (mid-case, high-renewable energy cost, low-battery cost, low-renewable energy cost, and low-wind cost) (Cole et al., 2020) (Gagnon et al., 2020). The low-renewable cost scenario forecasts that renewables will provide approximately 70% of electricity produced in the US by 2050 (Cole et al., 2020). Of the five scenarios NREL modeled, this scenario is most compatible with the goal of net zero greenhouse gas emissions economywide by 2050, although a zero net regulatory mandate by 2050 or earlier would result in even lower power sector emissions.

NREL provides several emissions metrics associated with each scenario. The metric used for this analysis was the "long-run marginal emissions rates for a region's load," which the Cambium documentation states is the appropriate metric to apply when estimating the impacts of load increases from "electric-sector emissions that would be induced by increased electric vehicle charging or by replacing a natural gas furnace with a heat pump" (Gagnon et al., 2020). While the emission forecast varies by year and by hour, the average marginal emissions over each

Long-Run Marginal End-Use Emissions Forecast Averaged over 15-years [kg CO_2/MWh]

		Pacific			Rocky Mountains						Southwest				Midwest										
	State	CA	OR	WA	CO	ID	MT	NV	WY	UT	AZ	OK	NM	TX	IL	IN	IA	KS	OH	MI	MN	MO	NE	ND	SD	WI
15-Year Period	2020-2035	104	117	122	390	209	249	225	485	451	324	321	438	336	271	435	368	282	423	334	272	382	347	434	391	377
	2022-2037	85	105	108	331	178	223	195	445	378	261	258	372	297	242	394	325	291	388	290	232	335	300	361	319	352
	2024-2039	80	88	91	333	177	228	181	439	396	225	254	311	285	231	368	319	337	363	308	225	330	289	343	290	331
	2026-2041	76	91	93	307	152	189	162	402	363	209	211	278	257	211	325	309	333	324	283	209	287	276	283	224	294
	2028-2043	74	93	95	296	160	177	161	393	366	203	161	253	242	209	310	288	308	293	301	181	257	262	255	181	268
	2030-2045	67	94	96	308	158	176	145	400	362	195	130	246	241	211	294	313	312	277	310	201	262	262	274	181	258
	2032-2047	59	93	91	287	147	162	141	380	332	189	119	236	240	199	274	335	326	268	294	222	256	268	294	168	263
	2034-2049	57	97	90	309	132	151	126	368	290	180	96	219	233	213	261	331	339	252	273	213	254	263	274	139	283

		Northeast											Southeast										
	State	CT	DE	MA	MD	ME	NH	NJ	NY	PA	RI	VT	AL	AR	FL	GA	KY	LA	MS	NC	SC	TN	VA	WV
15-Year Period	2020-2035	144	229	100	224	147	43	169	136	333	100	35	481	333	348	342	439	329	369	311	313	323	150	343
	2022-2037	123	203	74	205	130	35	157	108	312	81	4	456	303	325	336	380	292	334	292	286	269	115	313
	2024-2039	122	209	69	215	128	33	165	111	317	81	3	454	302	290	298	382	282	333	279	246	288	123	273
	2026-2041	115	195	72	204	117	29	154	105	278	83	4	419	268	247	244	332	262	303	250	218	248	123	238
	2028-2043	114	179	71	199	113	22	167	114	240	80	7	407	249	226	229	291	253	275	229	206	236	108	224
	2030-2045	111	189	73	208	109	21	186	122	250	87	9	404	242	225	223	278	258	270	238	211	252	110	232
	2032-2047	107	181	60	193	101	22	186	118	230	84	10	400	248	233	214	296	268	276	229	211	250	103	241
	2034-2049	113	172	65	205	99	27	178	121	210	87	11	365	238	210	181	299	256	259	211	214	250	107	240

Legend - Average kg CO_2/Mwh End-Use

100 200 300 400 500

Fig. 3. Average 15-year long-run marginal end-use emissions forecasted for NREL's low-cost renewable energy scenario.

15-year period analyzed is shown by state in Fig. 3. Average marginal emissions rates vary widely by state with a range of 3–485 kg CO_2 emitted per MWh of end-use electricity consumed. As expected, the emissions forecast generally improves over time, although some variation is observed due to complexities of the Cambium analysis for which the forecast is affected by many variables including renewable energy policy implementation, power plant construction and retirement, shifting economics, and changes in technology performance over time (Gagnon et al., 2020).

The annual CO_2 emissions from each heating system in each city simulated was calculated from the sum of the products of the hourly electricity used and the long-range marginal end-use emission rate for the state in which the city was located (Equation (1)) for each hour.

Equation (1) – Annual CO_2 Emissions from End-Use Electricity

$$m_{CO2_elec_annual}\ [kg\ C0_2] = \sum_{h=1}^{8760}\left(E_h\ [kWh] \times\ co2_lrmer_enduse_h\left[\frac{kg\ C0_2}{kWh}\right]\right)$$

where E_h was the modeled electricity consumed by the heat pump (sum of compressor, electric resistance, and air handler fan) or furnace air handler fan each hour and $co2_lrmer_enduse_h$ was the hourly emissions for each unit of end-use electricity consumed for that hour for the state in which the city was located.

Lifetime emissions were estimated by summing the annual CO_2 emissions over the median service life of the equipment (15-year period), starting from the date of the heat pump or gas furnace installation.

2.3.2. Carbon dioxide emissions from furnace natural gas combustion

The annual CO_2 emissions from combustion of natural gas by the modeled furnace in each city was calculated by multiplying the annual natural gas consumption by an emission factor of 2.445 kg CO_2 per Joule of natural gas burned (U.S. Energy Information Adminstration, 2020). Lifetime emissions were estimated by summing the annual CO_2 emissions over a 15-year period, starting from the date of gas furnace installation.

2.3.3. Global warming potential impact of refrigerant leakage

In homes with existing air conditioning, replacing a gas furnace with a heat pump is expected to increase the refrigerant charge of the system and the potential for refrigerant emissions due to additional refrigerant containing components (e.g. extra piping, valves, accumulator). No published literature was found that estimated the difference in the amount of refrigerant for a residential air conditioner versus heat pump. An informal survey of manufacturer product data for both air conditioners and heat pumps found widely variable differences in refrigerant charge per unit of rated cooling capacity across manufacturer model lines, suggesting that equipment is generally not designed to minimize refrigerant charge. Generally, refrigerant charge increased with increasing system efficiency, but the difference in charge between air conditioners and heat pumps decreased with increased system efficiency. The average value of R410-A refrigerant charge for three current models of air conditioners (Trane 4TTR4, 4TTR6, and 4TTA7) (Trane Technologies, 2021) and heat pumps (Trane 4TWR4, 4TWR6, 4TWR7) (Trane Technologies, 2021) for three common cooling capacities in residential homes (10.5, 14, 17.5 kW) was calculated. The average air conditioner charge was 0.28 kg/kW and the average heat pump charge was 0.34 kg/kW. Thus, an increase of 0.06 kg/kW refrigerant charge was included in the heat pump analysis.

When adding a heat pump in place of a furnace/air conditioner combination unit, the only incremental increase in refrigerant is due to the modest increase in refrigerant attributed to the heat pump function, rather than cooling-only, design. In addition to accounting for the refrigerant increase for a heat pump versus an air conditioner, replacing a gas furnace with a heat pump in a house where there is no previous air conditioning increases the amount of refrigerant in use and the potential for refrigerant emissions. Accounting for this additional refrigerant use required estimating the fraction of homes without existing refrigerant-based air conditioning. Data from 2015 published by the US Energy Information Administration (EIA) (U.S. Energy Information Administration, 2015b) was used to estimate the penetration of air conditioning equipment by IECC climate region. A quadratic regression using the least squares method was applied to EIA data of air conditioning adoption rates in homes in the South region from 1980 to 2009 (U.S. Energy Information Administration, 2011) to forecast a future adoption curve for all regions through 2036 (Fig. 4). The data from the South region was used because it provided historical data of an adoption curve reaching nearly 100% saturation. This projected adoption curve reflects historic overall trends of air conditioner adoption, which are likely to continue and perhaps intensify as the climate warms and heat waves become more severe.

Estimated refrigerant (*Ref*) leakage amount over a 15-year lifetime was based on the capacity of the heat pump as shown in Equation (2).

Equation (2) – Lifetime refrigerant emissions

$$m_{Ref_lifetime}\ [kg] = \left(0.06\ \frac{kg}{kW} + 0.28\left[\frac{kg}{kW}\right] \times Cap\ [kW]\ \times (1 - f_{ac})\right)\ \times f_e$$

where 0.06 kg/kW is the additional refrigerant for a heat pump, 0.28 kg/kW is the refrigerant in an air conditioner, *Cap* is the nominal cooling capacity of the system for each city as determined by the EnergyPlus autosize calculation, f_{ac} is the fraction of homes with existing air conditioning for the region in which each city is located (Fig. 4), and f_e is the fractional emissions from the heat pump system over the 15-year lifetime based on the year of installation. The fractional emissions, f_e, is estimated from Equation (3).

Equation (3) – Lifetime refrigerant emissions ratio

$$f_e\ [-] = 1.125e^{-0.045(y-2020)}$$

where y is the heat pump installation year. The equation represents a simplified leak model where, for a heat pump installed in the year 2020, 112.5% of the initial refrigerant charge is emitted over its lifetime from installation, service (including leak repairs and re-charge), and disposal, for an annualized average of 7.5% over a 15-year life. This estimate is consistent with the range of refrigerant emission rates reported by the US Environmental Protection Agency (EPA) (United States Environmental Protection Agency, 2020), assuming emissions of 0.5% at installation, 5.8% annually each year for 15 years, and 25% at disposal. Additionally, Equation (3) models an 80% reduction in lifetime emissions every five years (based on the date of heat pump installation) expected to occur as a result of the American Innovation and Manufacturing Act, which directs the EPA

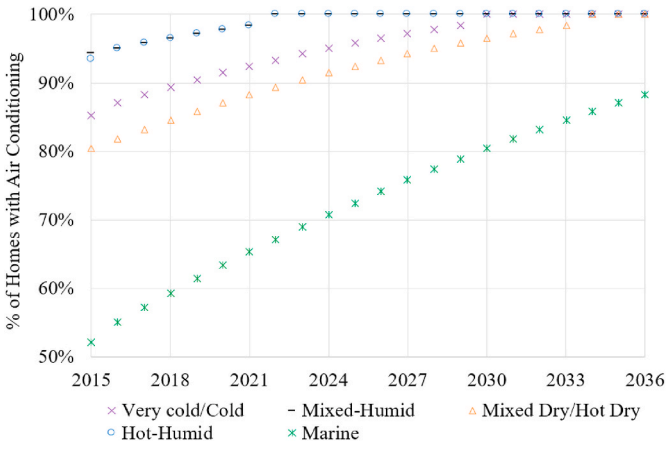


Fig. 4. Forecasted residential air conditioning adoption rates by IECC climate region based on data from the US EIA in 2015 and historical adoption curve from the Southern US from 1980 to 2009.

to reduce the environmental impacts of hydrofluorocarbons (U.S. Environmental Protection Agency, 2021). This includes a national phasedown, new regulations to reduce fugitive and end-of-life refrigerant emissions to support this phasedown, and facilitation of a transition to next-generation refrigerants with lower GWP.

The 20-year and 100-year GWP were used to calculate the impact of refrigerant emissions compared to the CO_2-eq over these two relevant time scales. The mass of the leaked refrigerant was multiplied by the appropriate GWP for the heat pump refrigerant used. For simulation years 2022 and 2024, the heat pump was assumed to contain R-410A, which has a 20-year GWP of 4260 CO_2-eq and a 100-year GWP of 1924 CO_2-eq (Myhre et al., 2013). In expectation of a national phasedown of R-410A (U.S. Environmental Protection Agency, 2021), for installation years of 2026 and beyond, the heat pump was assumed to use R-32, which has a 20-year GWP of 2430 CO_2-eq and a 100-year GWP of 677 CO_2-eq (Myhre et al., 2013).

2.3.4. Global warming potential impact of natural gas leakage

Fugitive methane emissions from natural gas production sites are the second greatest contributor to the GHG footprint of the US after combustion CO_2 (Alvarezet al, 2018). A recent study aggregated emission data from basin level production, consumption, trade and spatial evaluation of pipeline connections between the basin and consumer states, and estimated that the production stage methane emissions vary by state from 0.9% to 3.6% of end use consumption, where emissions were highest in the Southwest, Pacific, Midwest and Rocky Mountains and lowest in the Southeast and Northeast (Burns and Grubert, 2021).

To accurately predict the methane emissions from the operation of a heat pump, the current study included the production stage fugitive methane emissions attributable to the electricity generated in natural gas power plants, which decreases over time as more renewables penetrate the US grid. NREL's Cambium tool forecast includes the breakdown of generation type by hour, including the generation provided by three types of natural gas power plants (Table 3). The efficiency of natural gas power plant varies with installation year, technology, and process used for generation. Average efficiencies derived from recent literature (Damen et al., 2006) (Mom, 2013) (Manzolini and Jansen, 2013) (U.S. Department of Energy, 2016) for the three types of natural gas power plants (Table 3) were applied to the Cambium data to calculate the methane consumed for power generation for electricity consumed in each scenario (heating system type, state, and year of installation). Subsequently, the production stage methane emissions attributable to electricity consumed was calculated for each city based on the emissions (as a percent of end use consumption) by state reported by Burns et al. (Burns and Grubert, 2021). Losses associated with transmission and distribution of natural gas were not included in the current analysis due to the fact that natural gas pipelines will remain pressurized for other applications and will likely not be impacted by incremental reductions in generation and consumption.

For the gas furnace, the heating energy consumed over the 15-year period in each city (in Joules) was converted to mass using the heating value for methane of 50 MJ/kg (Moran et al., 2014). Subsequently, the production stage methane emissions attributable to natural gas consumed by the furnace was calculated for each city based on the emissions (as a percent of end use consumption) by state reported by Burns et al. (Burns and Grubert, 2021). In addition to the fugitive methane emissions from production, the current analysis also accounted for behind the meter (BTM) leakage for the natural gas consumed in the furnace. BTM emissions from home distribution system can be estimated at 0.5% of end use consumption based on an estimate in a report by the California Energy Commission (Fischer et al., 2018). The methane emissions were multiplied by 84 to determine the 20-year GWP CO_2-eq and by 28 to determine the 100-year GWP CO_2-eq (Myhre et al., 2013). The analysis assumes that all BTM methane emissions associated with natural furnace consumption are eliminated with a heat pump replacement.

BTM leaks, which are estimated at 0.5% of consumption (Fischer et al., 2018), are attributed to incomplete combustion, pilot light operation, and leaks within the home's distribution system. Leaks in the home's distribution system are not expected to be reduced with removal of a furnace if the distribution system remains pressurized to serve other natural gas appliances (unless the portion of the distribution system serving the furnace can be removed). Therefore, a fraction of BTM leakage attributed to gas furnaces could remain with installation of a heat pump, however, this is small in context of the GWP impacts associated with natural gas production leakage.

Table 3
Modeled natural gas power plant efficiencies.

Natural Gas Power Plant Type	Modeled Average Efficiency
Combined Cycle	58%
Combined Cycle with Carbon Capture and Storage	49%
Combustion Turbine	35%

3. Results and discussion

The CO_2 emissions associated with end-use heating electricity consumption in buildings with a variable speed heat pump is impacted by the fraction of heating energy provided by the heat pump mode (COP range of 1.3–4.7) versus the electric resistance heat strips (COP of 1). The population-weighted average of the simulation results for the fraction of heating energy provided and electricity consumed by the heat pump mode is shown by region in Fig. 5. Overall, for the population-weighted average across the 99 cities simulated, 87% of the heating energy delivered and 68% of the heating electricity consumed was by the heat pump mode, with the remainder being met by the electric resistance heat strip. Variations among region were observed, where climates with lower outdoor air temperatures (Midwest, Northwest) generally had increased reliance on electric resistance strip heat due to the switchover to heat strips below an outdoor-air temperature of −18 °C. The results also indicated that percent of the load met by the heat pump mode was approximately the same in both construction year models. This is because with more energy efficient construction the heat pump size scales down with the design day cooling load so the heat strips are in use for a similar fraction of the heating hours.

In practice, the actual fraction of heating provided by the heat pump mode is highly dependent upon how the heat pump is operated, whether or not setbacks are used by the homeowner, and the time allowed by the control system to recover the home temperature using the heat pump alone. A study by Larson et al. (2013), monitoring six residential heat pumps in the Redmond, Oregon area over 35 days, determined that the average heating electricity consumed by the heat pump mode was 72% (range 36–94%), which compared well with 71–72% projection for the same city simulated in the current study. The wide range of results for the load met by the heat pump mode reported in the field study highlights the importance of implementing heat pump controls that prevent operation of the electric resistance strip heat when the load can be met by the heat pump alone to achieve maximum potential in GHG emission reductions.

The US population-weighted average forecasted 20-year and 100-year GWP emissions over the 15-year life of the heating equipment are shown in Fig. 6 for both the gas furnace (GF) and heat pump (HP) installed in three future years: 2022, 2028, and 2036 for both construction vintages: 2006 in the left column and 2018 in the right column. Reductions in CO_2, 20-Year GWP and 100-Year GWP emissions were consistently reduced by 47–48% from 2006 to 2018 as a result of the improved building envelope, highlighting the importance of overall building efficiency in reducing emissions resulting from residential space heating. The CO_2 emissions for gas furnaces in the same home remain nearly constant for the 2022–2036 installation years, except for a small reduction over time in CO_2 emissions associated with electricity

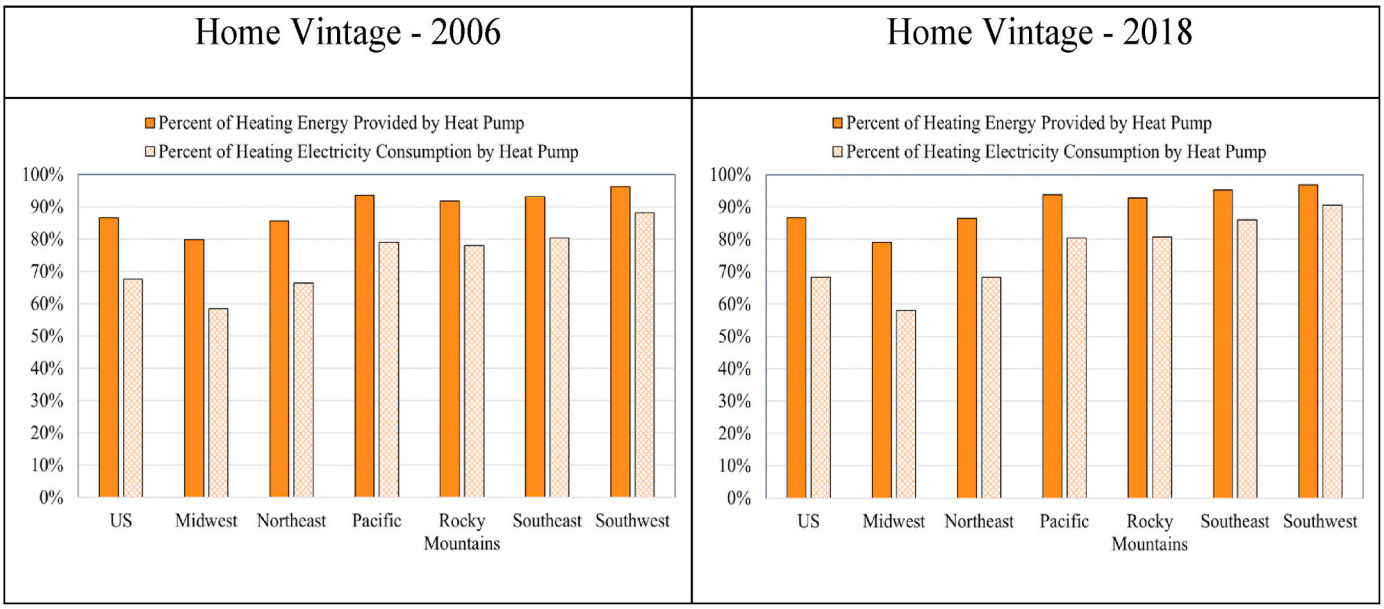


Fig. 5. –Percent of heating energy and electricity consumed by the heat pump mode by region in the residential home for 2006 construction (left) and 2018 construction (right).

use of the AHU fan. The CO_2 emissions for the heat pump electricity used by AHU fan, compressor, and electric resistance heat strips decrease over time due to forecasted reductions in the long-run marginal CO_2 emissions from electricity generation. It is interesting to note that the AHU fan for the heat pump generates slightly more emissions compared to the AHU for the furnace due to longer runtimes as a result of the lower supply air temperatures delivered by the heat pump.

The US population-weighted average forecasted 20-year and 100-year GWP equivalent CO_2 impacts from methane emissions associated with the natural gas used by the furnace (from natural gas production and BTM leakage) are more significant, about 3–8 times, than the comparable emissions associated with heat pumps due to natural gas production leakage for natural gas power plant operation and refrigerant emissions (Fig. 6). The difference is more significant in the 20-Year GWP analysis than in the 100-Year GWP analysis because methane's global mean atmospheric lifetime is estimated to be 8.4 years (Ehhalt et al., 2018).

The results in Fig. 6 are converted to a percent reduction in CO_2, 20-Year, and 100-Year GWP emissions for replacement of a natural gas furnace with a heat pump by region based on year of installation and for two home vintages (2006 and 2018) (Fig. 7). Although the absolute reduction in emissions changes significantly by year of home construction (Fig. 6), the percent reduction in emissions are nearly identical because the heating system is sized to meet the loads of the respective building construction. This result suggests that specific details of home construction do not significantly impact the expected percent emissions reduction resulting from replacement of a gas furnace with a heat pump, although energy efficiency of the home significantly impacts absolute emissions and energy costs. Overall, from the population weighted average of 99 cities simulated in the US, emission reductions resulting from replacing a gas furnace with a heat pump were forecasted to be 38–53% for CO_2, 53–67% for 20-Year GWP, and 44–60% for 100-Year GWP, with reductions increasing over time.

Analyzing results by region showed that the greatest reduction in emissions from heat pump installation is expected in the Pacific and Northeast regions (Fig. 7) where long-range marginal emissions from electricity generation were forecasted to be the lowest (Fig. 3). In the Pacific region, which had the greatest forecasted reductions, emission reductions were forecasted to be 76–81% for CO_2, 66–83% for 20-Year GWP, and 68–83% for 100-Year GWP, with reductions increasing over time. Several studies focused on CO_2 emissions in California can be compared to the current result. Brockway and Delforge, 2018 predicted a 46–54% reduction in annual CO_2 emissions for electrification of space heating in California homes using the short-run marginal emissions projected for 2030. Tarroja et al. (2018) predicted 30–40% reduction in CO_2 emissions in California from electrification of space and water heating in the year 2050. Raghaven et al., 2017 predicted about 75% reduction in average annual CO_2 emissions in California for replacement of an 82% efficient natural gas water heater with a heat pump in 2036. Overall, the increased CO_2 reductions found in the current study in comparison to these other studies can be attributed to the current analysis covering the product lifetime, accounting for fugitive emissions, and applying the recently published Cambium long-range marginal emission rates. A comprehensive study by Mahone et al. (2018) considering fugitive methane and refrigerant emissions predicted a 76–88% 100-year GWP emissions reduction resulting from residential electrification in California in 2050, which compares well with 81–83% 100-year GWP reduction predicted in the current study for a heat pump operating from 2036 to 2050 in the Pacific region.

The Rocky Mountain, Southeast, and Southwest regions had mid-range results where emission reductions were forecasted to be 32–63% for CO_2, 48–72% for 20-Year GWP, and 41–62% for 100-Year GWP, with reductions increasing with time. The region with the lowest reductions was the Midwest, which has higher long-range marginal emissions (Fig. 3) in combination with the greatest use of electric resistance strip heat (Fig. 5). In the Midwest region, emission reductions were forecasted to be 13–31% for CO_2, 45–57% for 20-Year GWP, and 28–43% for 100-Year GWP, with reductions increasing with time.

4. Conclusions and policy implications

This study provided first detailed analysis comparing the expected 15-year operational GHG emission impacts (2022–2050) from a natural gas furnace and a heat pump for heating a single-family residential home in six regions across the US. The methodology used in this analysis considered the time varying nature of the electricity generation mix by simulating the hourly heating loads of the house in various climate zones. The models of the heating equipment consisted of a 96% AFUE furnace and variable speed heat pump (COP of 3.0) with backup electrical resistance heaters. The hourly heating loads were matched with

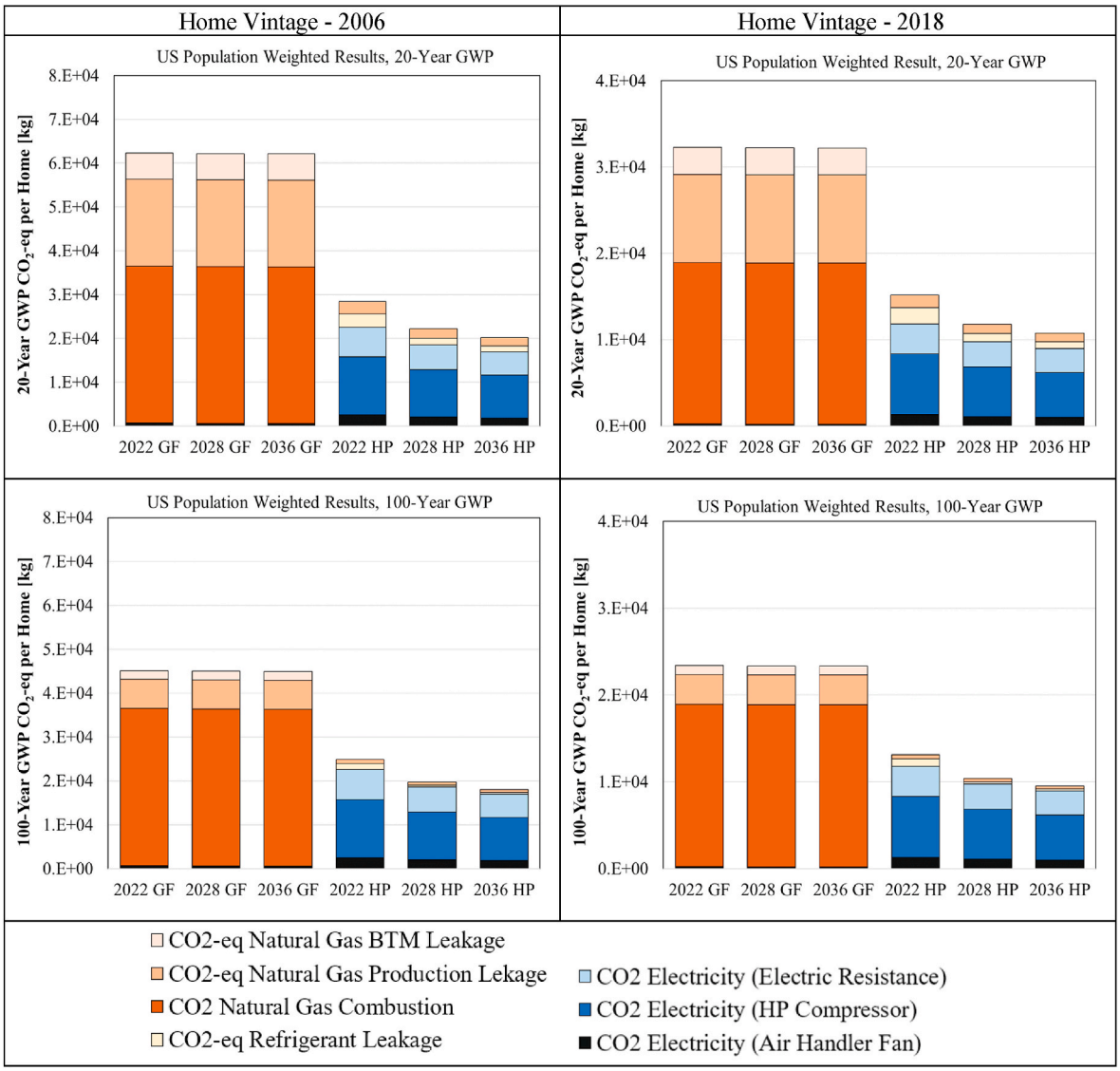


Fig. 6. US population-weighted average forecasted GWP20 and GWP100 emissions over the 15-year life of the heating equipment for both the gas furnace (GF) and heat pump (HP) installed in three future years: 2022, 2028, and 2036.

the respective long-run marginal emissions for the end-use electricity consumed. The analysis also accounted for other significant GHG emissions associated with gas furnaces and heat pumps: CO_2 emissions from natural gas combustion in homes, emissions from methane leaks during production and behind the meter in homes, and emissions from refrigerant leaks. These fugitive methane and refrigerant leaks were factored into the analysis using the CO_2-eq of the GWP of these gases on a 20-year and 100-year timescale. The study did not compare the embodied carbon and cost assessment of each of the equipment.

The results from this study support that residential heat pump installation in place of natural gas furnaces will significantly reduce operational GHG emissions under a wide range of geographic and climatic conditions. Policies to encourage greater residential heat pump penetration with the aim of reducing GHG emissions are supported by the results of this study. Regional results calculated based on population weighted results from 99 cities may not be representative of smaller cities with extremely cold weather. In comparing the results of homes representative of 2006 and 2018 construction practices, this study concluded that the percentage GHG emission reductions from the replacement of a gas furnace with an electric heat pump are not sensitive to home construction parameters.

Prior work on GHG emission impacts of residential electrification did not consider methane leaks and refrigerant leaks and have been limited to a specific geographic location (mostly California). Methane emissions from natural gas production sites are the second most significant source of GHG impacts from residential natural gas furnaces and policies supporting heat pump deployment will have the direct benefit of eliminating those emissions. Emissions from transportation, storage, and local distribution systems are assumed to remain and were not considered here. Emissions of refrigerants contribute only modestly to the results in this work because only the additional refrigerant associated with heat pumps over and above refrigerant use already projected for use in air conditioning was considered. This result should not, however, be interpreted as suggesting that refrigerant emissions overall are not impactful. Indeed, they are very significant – about 170 MMT CO_2-eq in 2019 (United States Environmental Protection Agency, 2021)– and their use and production is being phased down in the US and abroad according to the Kigali Amendment to the Montreal Protocol (United Nations Treaty Collection, 2019). In addition, lower-GWP refrigerant candidates to replace R-410A are expected to work equivalently well in residential air conditioning-only and heat pump applications. For this reason, policies to expand heat pump deployment will only modestly impact refrigerant emissions from the residential sector, a change

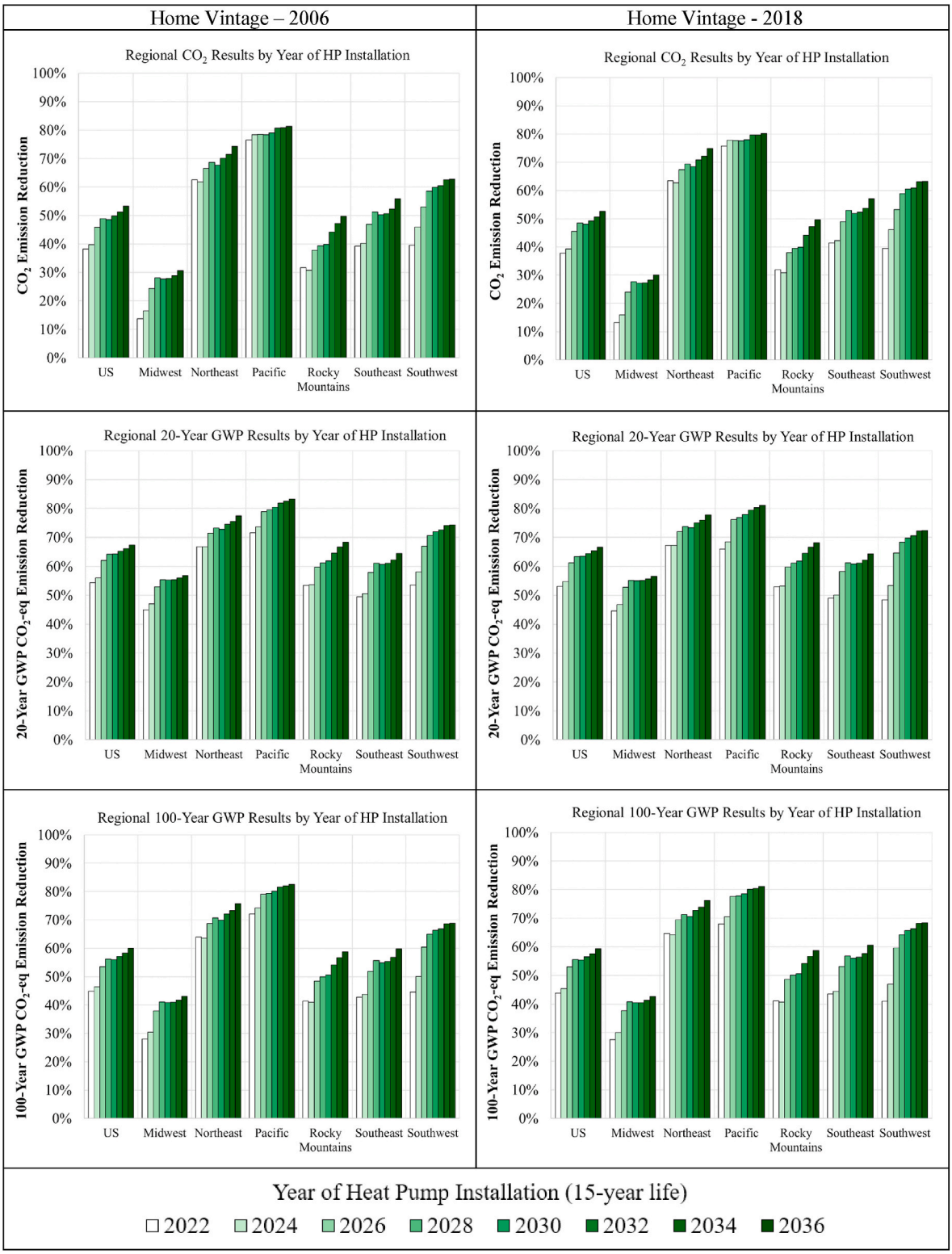


Fig. 7. Regional population-weighted average results for emission reductions by year of heat pump installation for CO_2 (top), 20-year GWP (middle), and 100-year GWP (bottom) for 2006 (left) and 2018 (right) construction years.

that is far outweighed by the significant reductions in carbon dioxide and methane emissions gained by deploying increasing numbers of heat pumps.

The NREL Cambium low renewable energy cost grid scenario used to predict the electricity grid in the future reflects that the 70% of US electricity will come from clean sources in 2050. There have been new targets set by the recent administration and initiatives by the department of energy to reach 100% clean electricity by 2035 (Department of Energy, 2021). In such a dramatic shift in policies, it is likely that even greater reductions in GHG emissions will be achieved by heat pump installations than the numbers predicted in this study. In addition, heat pump load flexibility controls that could automatically shift a portion of heating loads to times when low-carbon intensity

electricity is available could further increase emission reductions.

Biogas from organic waste is growing rapidly in the US and a portion of it is predicted to be used in electricity generation by the Cambium data. Biogas was not considered as a source of methane for end-use natural gas furnace consumption. At present landfills account for 90% of the US biogas production and incorporating less concentrated sources such as animal manure and forestry residues face more economic barriers and technology hurdles (Gasper and Searchinger, 2018). Studies have predicted 4%–10% of the natural gas consumed in the US can potentially be replaced by biogas (Cyrs and Feldmann, 2021). However, as the US works to reduce emissions, it is likely that industrial needs that are harder to electrify would be the primary consumer of available biogas. Thus, this study did not consider biogas as a potential fuel source for residential building sector furnaces, where electrification replacements are available.

This study aimed to answer the question affecting energy policy choices in electrification of residential space heating by comprehensive analysis of the GHG impacts of gas furnace and heat pump. Accelerated heat pump deployment policy has a prominent potential for reducing GHG emissions from residential buildings in the US.

Data availability

Please see https://wcec.ucdavis.edu/analysis-of-greenhouse-gas-emissions-from-residential-heating-technologies-in-usa/for information on data availability.

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Credit authorship contribution statement

Theresa Pistochini: conducted the literature review, determined all modeling parameters and methods, reviewed results and wrote the publication. **Mitchal Dichter:** programmed the EnergyPlus model and script to run and post-process several hundred simulations. **Subhrajit Chakraborty:** conducted the literature review, determined all modeling parameters and methods, reviewed results and wrote the publication. **Nelson Dichter:** developed modeling methodology. **Aref Aboud:** supported the team as a student assistant.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Locations and weather files simulated to produce regional and US results (weighted averages by county population).

State	City	County	WMO Station Number	County Population	IECC Climate Zone	Building America Climate Zone	Region
IA	Des-Moines	Des Moines	725460	38,967	5	Cold	Midwest
IA	Sioux-City	Woodbury	725570	103,107	6	Cold	Midwest
IL	Belleville	St Clair	724338	89,512	4	Mixed-Humid	Midwest
IL	Chicago	Cook	725340	5,150,233	5	Cold	Midwest
IN	Evansville	Vanderburgh	724320	181,451	4	Mixed-Humid	Midwest
IN	Indianapolis	Marion	724380	964,582	5	Cold	Midwest
KS	Hays	Ellis	724518	28,553	5	Cold	Midwest
KS	Wichita	Sedgwick	724500	516,042	4	Mixed-Humid	Midwest
MI	Detroit	Wayne	725375	1,749,343	5	Cold	Midwest
MI	Houghton	Roscommon	726380	24,019	7	Very Cold	Midwest
MI	Traverse-City	Grand Traverse	726387	93,088	6	Cold	Midwest
MN	Duluth	St Louis	727450	199,070	7	Very Cold	Midwest
MN	Minneapolis	Hennepin	726580	1,265,843	6	Cold	Midwest
ND	Bismark	Burleigh	727640	95,626	6	Cold	Midwest
ND	Fargo	Cass	727530	181,923	7	Very Cold	Midwest
NE	Omaha	Douglas	725530	571,327	5	Cold	Midwest
OH	Cincinnati	Hamilton	724297	817,473	4	Mixed-Humid	Midwest
OH	Columbus	Franklin	724280	1,316,756	5	Cold	Midwest
SD	Yankton	Yankton	726525	22,814	5	Cold	Midwest
SD	Sioux-Falls	Minnehaha	726510	193,134	6	Cold	Midwest
WI	Milwaukee	Milwaukee	726400	945,726	6	Cold	Midwest
WI	Rhinelander	Oneida	727415	35,595	7	Very Cold	Midwest
CT	Bridgeport	Fairfield	725040	943,332	5	Cold	Northeast
MA	Boston	Suffolk	725090	803,907	5	Cold	Northeast
ME	Portland	Cumberland	726060	295,003	6	Cold	Northeast
ME	Presque-Isle	Aroostook	727130	67,055	7	Very Cold	Northeast
NH	Concord	Merrimack	726050	151,391	6	Cold	Northeast
NH	Manchester	Hillsborough	743945	417,025	5	Cold	Northeast
NJ	Newark	Essex	725020	798,975	4	Mixed-Humid	Northeast
NJ	Trenton	Mercer	724095	367,430	5	Cold	Northeast
NY	Buffalo	Erie	725280	918,702	5	Cold	Northeast

(*continued on next page*)

(*continued*)

State	City	County	WMO Station Number	County Population	IECC Climate Zone	Building America Climate Zone	Region
NY	New-York	New York	744860	1,628,706	4	Mixed-Humid	Northeast
NY	Utica	Oneida	725197	228,671	6	Cold	Northeast
PA	Bradford	McKean	725266	40,625	6	Cold	Northeast
PA	Philadelphia	Philadelphia	724080	1,584,064	4	Mixed-Humid	Northeast
PA	Pittsburgh	Allegheny	725200	1,216,045	5	Cold	Northeast
RI	Providence	Providence	725070	638,931	5	Cold	Northeast
VT	Burlington	Chittenden	726170	163,774	6	Cold	Northeast
CA	Bishop	Inyo	724800	18,039	5	Cold	Pacific
CA	Crescent City	Del Norte	725946	27,812	4	Marine	Pacific
CA	Imperial	Imperial	747185	181,215	2	Hot-Dry	Pacific
CA	Los-Angeles	Los Angeles	722950	10,039,107	3	Marine	Pacific
CA	Sacramento	Sacramento	724839	1,552,058	3	Hot-Dry	Pacific
CA	San-Jose	Santa Clara	724945	1,927,852	3	Marine	Pacific
OR	Portland	Multnomah	726980	812,855	4	Marine	Pacific
OR	Redmond	Deschutes	726835	197,692	5	Cold	Pacific
WA	Seattle	King	727930	2,252,782	4	Marine	Pacific
WA	Spokane	Spokane	727850	522,798	5	Cold	Pacific
CO	Alamosa	Alamosa	724620	16,233	6	Cold	Rocky Mountains
CO	Aspen	Pitkin	724676	17,767	7	Very Cold	Rocky Mountains
CO	Denver	Denver	725650	727,211	5	Cold	Rocky Mountains
CO	Trinidad	Las Animas	724645	14,506	4	Cold	Rocky Mountains
ID	Boise	Ada	726810	481,587	5	Cold	Rocky Mountains
ID	Idaho-Falls	Bonneville	725785	119,062	6	Cold	Rocky Mountains
MO	Kansas-City	Jackson	724463	703,011	4	Mixed-Humid	Rocky Mountains
MO	St.-Joseph	Buchanan	724490	87,364	5	Cold	Rocky Mountains
MT	Billings	Yellowstone	726770	161,300	6	Cold	Rocky Mountains
NV	Las-Vegas	Clark	723860	2,266,715	3	Hot-Dry	Rocky Mountains
NV	Reno	Washoe	724880	471,519	5	Cold	Rocky Mountains
UT	Salt-Lake-City	Salt Lake	725720	1,160,437	5	Cold	Rocky Mountains
UT	St.-Geroge	Washington	724754	129,375	3	Hot-Dry	Rocky Mountains
UT	Vernal	Uintah	725705	35,734	6	Cold	Rocky Mountains
WY	Cheyenne	Laramie	725640	99,500	6	Cold	Rocky Mountains
WY	Jackson	Teton	725776	23,464	7	Very Cold	Rocky Mountains
AL	Birmingham	Jefferson	722280	658,573	3	Mixed-Humid	Southeast
AL	Mobile	Mobile	722230	413,210	2	Hot-Humid	Southeast
AR	Fayetteville	Washington	723445	239,187	4	Mixed-Humid	Southeast
AR	Little-Rock	Pulaski	723405	391,911	3	Mixed-Humid	Southeast
DE	Wilmington	New Castle	724089	558,753	4	Mixed-Humid	Southeast
FL	Jacksonville	Duval	722060	957,755	2	Hot-Humid	Southeast
FL	Miami	Miami-Dade	722020	2,716,940	1	Hot-Humid	Southeast
GA	Atlanta	Fulton	722190	1,063,937	3	Mixed-Humid	Southeast
GA	Rome	Floyd	723200	98,498	4	Mixed-Humid	Southeast
GA	Savannah	Chatham	722070	289,430	2	Hot-Humid	Southeast
KY	Louisville	Jefferson	724235	766,757	4	Mixed-Humid	Southeast
LA	New-Orleans	Orleans	722310	390,144	2	Hot-Humid	Southeast
LA	Shreveport	Bossier	722484	127,039	3	Hot-Humid	Southeast
MD	Baltimore	Baltimore	724060	827,370	4	Mixed-Humid	Southeast
MS	Gulfport	Harrison	747685	208,080	2	Hot-Humid	Southeast
MS	Jackson	Hinds	722350	231,840	3	Hot-Humid	Southeast
NC	Charlotte	Mecklenburg	723140	1,110,356	3	Mixed-Humid	Southeast
NC	Raleigh	Wake	723060	1,111,761	4	Mixed-Humid	Southeast
SC	Charleston	Charleston	722080	411,406	3	Hot-Humid	Southeast
TN	Memphis	Shelby	723340	937,166	3	Hot-Humid	Southeast
TN	Nashville	Davidson	723270	694,144	4	Hot-Humid	Southeast
VA	Virginia-Beach	Virginia Beach city	723080	449,974	4	Mixed-Humid	Southeast
WV	Charleston	Kanawha	724140	178,124	4	Mixed-Humid	Southeast
WV	Morgantown	Monongalia	724176	105,612	5	Cold	Southeast
AZ	Flagstaff	Coconino	723755	143,476	5	Cold	Southwest
AZ	Kingman	Mohave	723700	212,181	3	Hot-Dry	Southwest
AZ	Phoenix	Maricopa	722780	4,485,414	2	Hot-Dry	Southwest
AZ	Prescott	Yavapai	723723	235,099	4	Mixed-Dry	Southwest
NM	Albuquerque	Bernalillo	723650	679,121	4	Mixed-Dry	Southwest
NM	Las-Cruces	Doña Ana	722695	218,195	3	Hot-Dry	Southwest
NM	Santa-Fe	Santa Fe	723656	150,358	5	Cold	Southwest
OK	Oklahoma-City	Oklahoma	723544	797,434	3	Mixed-Humid	Southwest
TX	Dallas	Dallas	722590	2,635,516	3	Hot-Humid	Southwest
TX	Houston	Harris	722436	4,713,325	2	Hot-Humid	Southwest
TX	Lubbock	Lubbock	722670	310,569	4	Hot-Dry	Southwest

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